We breifly mentioned the series $\sum_{n=0}^{\infty} x^n$ before and saw it was equal to $\frac{1}{1-x}$ only for $x \in (-1,1)$

The series $\sum_{n=0}^{\infty} x_n$ is a specific Power Series and $(-1,1]$ is called the interval of convergence.

A general Power Series is of the form

$$\sum_{n=0}^{\infty} C_n X^n = C_0 + C_1 X + C_2 X^2 + C_3 X^3 + \ldots$$

where X is a variable and C_n are coefficients depending on n.

If we fix X to be some number, we can test the series for convergence. The values of X in which the series is convergent is called the interval of convergence.

The function $f(x) = C_0 + C_1 X + C_2 X^2 + \ldots$ has domain equal to the interval of convergence.

We can generalize more and discuss the series

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

which is called a power series centered at a.

Notice this series always converges at $x=a$.

We will spend the rest of this section finding intervals of convergence. This almost always uses the root and ratio test.

Ex:

What values of x does $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ converge?

Ratio test

$$a_n = \frac{(x-3)^n}{n} \qquad a_{n+1} = \frac{(x-3)^{n+1}}{n+1}$$

$$\lim_{n\to\infty} \left| \frac{\frac{(x-3)^{n+1}}{n+1}}{\frac{(x-3)^n}{n}} \right| = \lim_{n\to\infty} |x-3| \frac{n}{n+1} = |x-3|$$

we want $|x-3|<1$ so

$$-1<x-3<1$$

$$\underline{2<x<4}$$ is absolute Convergence

However, the series might converge if $x=2,4$ as at these values the ratio test is inconclusive,

If $x=2$, we have $\sum \frac{(-1)^n}{n}$ which converges
If $x=4$, we have $\sum \frac{1}{n}$ which diverges

So $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ converges for $x \in [2,4)$

Ex: $\sum_{n=0}^{\infty} n!\,x^n$

Ratio test: $a_n = n!\,x^n$, $a_{n+1} = (n+1)!\,x^{n+1}$

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(n+1)!\,x^{n+1}}{n!\,x^n}\right| = \lim_{n\to\infty} (n+1)|x|$$

$$\lim_{n\to\infty} (n+1)|x| = \begin{cases} 0 & \text{if } x=0 \\ +\infty & \text{if } x\neq 0 \end{cases}$$

So $\sum n!\,x^n$ converges only for $x=0$

Ex:

$$\sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$$

$$a_n = \frac{x^n}{(2n)!}, \quad a_{n+1} = \frac{x^{n+1}}{(2n+2)!}$$

$$\lim_{n\to\infty} \left| \frac{\frac{x^{n+1}}{(2n+2)!}}{\frac{x^n}{(2n)!}} \right| = \lim_{n\to\infty} |x| \left(\frac{(2n)!}{(2n+2)!} \right) = |x| \lim_{n\to\infty} \frac{(2n)!}{(2n+2)(2n+1)(2n)!}$$

$$= |x| \lim_{n\to\infty} \frac{1}{(2n+2)(2n+1)}$$

$$= 0 \quad \text{always}$$

So

$\sum \frac{x^n}{(2n)!}$ Converges for all $x \in \mathbb{R}$

For any Power Series, we have the following:

Thm:

For $\sum_{n=0}^{\infty} c_n (x-a)^n$ there are only three possibilities

1) The Series Converges only for $x=a$
2) The Series Converges for all $x \in \mathbb{R}$
3) There is a positive number R s.t. the Series Converges for $|x-a| < R$ and diverges if $|x-a| > R$

The value of R is called the radius of convergence

- In case 1 above $R=0$
- In case 2 above $R=\infty$

we can write $|x-a|=R$ as $x\in(a-R,a+R)$

At end points $x=a\pm R$, we need to test individually, as ratio/root test are always inconclusive here.

Ex:

$$\sum_{n=0}^{\infty}\frac{(-3)^n x^n}{\sqrt{n+1}}$$

$$a_n=\frac{(-3)^n x^n}{\sqrt{n+1}},\quad a_{n+1}=\frac{(-3)^{n+1}x^{n+1}}{\sqrt{n+2}}$$

$$\lim_{n\to\infty}\left|\frac{\frac{(-3)^{n+1}x^{n+1}}{\sqrt{n+2}}}{\frac{(-3)^n x^n}{\sqrt{n+1}}}\right|=\lim_{n\to\infty}\frac{3\sqrt{n+1}}{\sqrt{n+2}}|x|$$

$$=3|x|\lim_{n\to\infty}\sqrt{\frac{n+1}{n+2}}=3|x|$$

we want $3|x|<1$ so $|x|<\frac{1}{3}$, so radius is $\boxed{R=1/3}$

This means $\frac{1}{3} < x < \frac{1}{3}$. We now need to test end points.

$x = \frac{-1}{3}$, $\displaystyle\sum_{n=0}^{\infty} \frac{(-3)^n \left(\frac{-1}{3}\right)^n}{\sqrt{n+1}} = \sum \frac{1}{\sqrt{n+1}}$

<u>diverges</u> by comparison of p-series w/ $p = 1/2$

$x = \frac{1}{3}$

$$\sum \frac{(-3)^n \left(\frac{1}{3}\right)}{\sqrt{n+1}} = \sum \frac{(-1)^n}{\sqrt{n+1}}$$

<u>Converges</u> by Alternating Series test

So,

$\boxed{\text{Interval of Convergence is } \left(\frac{-1}{3}, \frac{1}{3}\right] \text{ with radius } R = \frac{1}{3}}$

Ex: $$\sum_{n=0}^{\infty} \frac{n(x+2)^n}{3^{n+1}}$$

$$a_n = \frac{n(x+2)^n}{3^{n+1}}, \quad a_{n+1} = \frac{(n+1)(x+2)^{n+1}}{3^{n+2}}$$

$$\lim_{n\to\infty} \left| \frac{\frac{(n+1)(x+2)^{n+1}}{3^{n+2}}}{\frac{n(x+2)^n}{3^{n+1}}} \right| = |x+2|\frac{1}{3} \lim_{n\to\infty} \frac{n+1}{n} = |x+2|\frac{1}{3}$$

want $|x+2|\frac{1}{3} < 1$

so $|x+2| < 3 \to -5 < x < 1$

Checking end points

$x=-5$ $\quad \sum \frac{n(-3)^n}{3^{n+1}} = \frac{1}{3}\sum (-1)^n n$ diverges

$x=1$ $\quad \sum \frac{n(3)^n}{3^{n+1}} = \frac{1}{3}\sum n$ diverges

So, Interval of Convergence is

$(-5, 1)$ with radius $R=3$